

Grade 7
Unit 10
Lesson 7 Practice

Name _____
Date _____
Period _____

1. Solve the equation $4(0.5p + 1.5) = 12$ using your preferred method.

2. Use substitution to verify your solution in question 1.

3. Solve the equation $\frac{5}{2} = \frac{1}{4}(z - 8)$ using your preferred method.

4. Use substitution to verify your solution in question 3.

5. Solve the equation $18 = \frac{3}{4}(36.4 - n)$ using your preferred method.

6. Use substitution to verify your solution in question 5.

7. The perimeter of a regular pentagon with equal sides is 12 inches. If the length of each side is $w + \frac{2}{5}$, the equation $5\left(w + \frac{2}{5}\right) = 12$ can be used to find the value of w . Determine the value of w .
8. The length of each side of the regular pentagon is $2\frac{2}{5}$ inches because $12 \div 5 = 2\frac{2}{5}$. Explain how the solution to the equation $5\left(w + \frac{2}{5}\right) = 12$ can be used to show this as well.

Sade was asked to solve the equation $\frac{2}{3}(b + 2.5) = 9.5$. Her work is shown.

$$\frac{2}{3}(b + 2.5) = 9.5$$

$$\frac{2}{3} \cdot b + \frac{2}{3} \cdot 2.5 = 9.5 \cdot \frac{2}{3}$$

$$\frac{2}{3}b + \frac{5}{3} = \frac{19}{3}$$

$$\frac{2}{3}b + \frac{5}{3} - \frac{5}{3} = \frac{19}{3} - \frac{5}{3}$$

$$\frac{2}{3}b = \frac{14}{3}$$

$$\frac{3}{2} \cdot \frac{2}{3}b = \frac{3}{2} \cdot \frac{14}{3}$$

$$b = \frac{14}{2} = 7$$

9. Explain where Sade made a mistake.
10. Determine the correct solution to the equation.